SUPPLEMENTARY MATERIALS: IMPROVING SKETCHING ALGORITHMS FOR LOW-RANK MATRIX APPROXIMATION VIA SKETCH-POWER ITERATIONS*

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Outline of the Supplementary Materials. In Section SM1, we illustrate the parameter guidance for TYUC17-SPI when $m \neq n$. In Section SM2, we show that TYUC17-SPI is insensitive to the test matrices, using Gaussian, Rademacher, Count-Sketch, and sparse random test matrices as examples. Section SM3 shows that the sparse Rademacher test matrices used in the main manuscript exhibits JL behavior empirically. In Section SM4, we recall the TYUC17-SPI variant in Section 2.7 and the mixed-precision version of TYUC17 mentioned at the end of Section 7, and compare their performance. In Section SM7, we show how to integrate SPI into the streaming algorithm of Tropp et al. [SM4], preliminarily study its empirical performance, and provide a proof idea for establishing the deviation bound.

SM1. Parameter Guidance of TYUC17-SPI when $m \neq n$. Table SM1 presents the parameter of TYUC17-SPI when $m = cn$ where $c \neq 1$. This is an extension of that in Section 2.6 in the main manuscript. The derivation is similar to those in Section B and is omitted.

Table SM1: Sketch size s selection in different spectrum decay; $r \leq s \leq T/(c+1)$

Flat	Poly ($\sigma_i = i^{-\alpha}$)			Exp ($\sigma_i = e^{-\alpha \cdot i}$)	
	$\alpha < \frac{1}{2}$	$\alpha \approx \frac{1}{2}$	$\alpha > \frac{1}{2}$	$\alpha < \frac{1}{2T}$	$\alpha \geq \frac{1}{2T}$
r	r	$\mathrm{Proj}_{[r, \frac{T}{c+1}]}\left(-\frac{\frac{T+c}{c+1}}{W\left(-\frac{T+c}{(c+1)ne}\right)} - 1\right)$	$\max\{r, \frac{(2\alpha-1)(T+3)-(c+1)}{2(c+1)\alpha}\}$	r	$\frac{T}{c+1}$

SM2. Insensitivity of the SPI to Test Matrices. Our parameter guidance is based on the assumption that the test matrices are Gaussian matrices. However, in practice, we can use other test matrices such as countsketches and sparse Rademacher matrices, which costs little extra storage to hold in memory. In Section 4 of the main manuscript we have used sparse Rademacher test matrices. Here we use a numerical example to show that TYUC17-SPI is insensitive to the test matrices.

EXAMPLE SM2.1. *In this example, we expand the experiment from numerical Example 4.1. We used the same synthetic data as in Section 4.1.1 but with different test matrices including Gaussian, CountSketch, sparse sign [SM2, Sect. 9.2], and sparse Rademacher matrices (*99% *sparsity). To be specific, we only show the case of* $q = 1$ *in SPI.*

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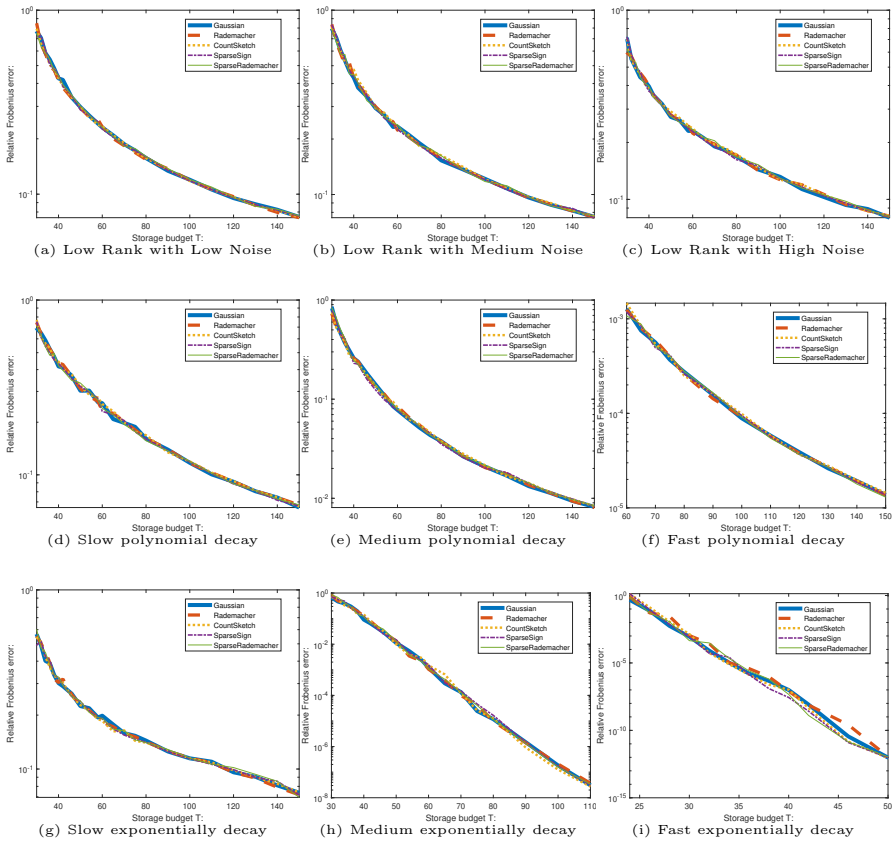

(a) Low Rank with Low Noise (b) Low Rank with Medium Noise (c) Low Rank with High Noise

(d) Slow polynomial decay (e) Medium polynomial decay (f) Fast polynomial decay

(g) Slow exponentially decay (h) Medium exponentially decay (i) Fast exponentially decay

Fig. SM1: Figures of Synthetic data. X-axis means the T represented the storage budget. Y-axis means the relative Frobenius error S_F. All results are averaged by 20 independent repeated experiments.

Figure SM1 shows that the results are almost identical for different test matrices except for perturbation by randomness when data matrix has more flatten spectrum decay. When the decay is more rapid, Gaussian embedding is slightly more stable. This observation allows us to use sparse matrices as test matrices in TYUC17-SPI in most cases, which costs little storage while succeeds the theoretical advantages for Gaussian matrices, as those obtained in Section 5 and 6. As a result, we can omit the storage cost of the test matrices in the derivation of the parameter guidance.

SM3. Numerical Experiments on JL Property of SparseRademacher Matrices. In this section, we provide numerical evidence to show that SparseRademacher matrices may satisfy the Johnson-Lindenstrauss (JL) property in high-dimensional spaces in practice. We conduct experiments to demonstrate that SparseRademacher matrices can effectively preserve the distances between points in high-dimensional space when projected into a lower-dimensional space. We test the JL property by checking the subspace embedding property [SM1, Prop. 5.2]:

PROPOSITION SM3.1 (c.f. Prop. 5.2 of [SM1]). *Suppose that* $\mathbf{U} \in \mathbb{R}^{n\times d}$ *is a matrix with orthonormal columns. For a matrix* $\mathbf{\Phi} \in \mathbb{R}^{s\times n}$*, the subspace embedding property is equivalent with the condition*

$$1-\varepsilon \;\le\; \sigma_{\min}(\mathbf{\Phi}\mathbf{U}) \;\le\; \sigma_{\max}(\mathbf{\Phi}\mathbf{U}) \;\le\; 1+\varepsilon.$$

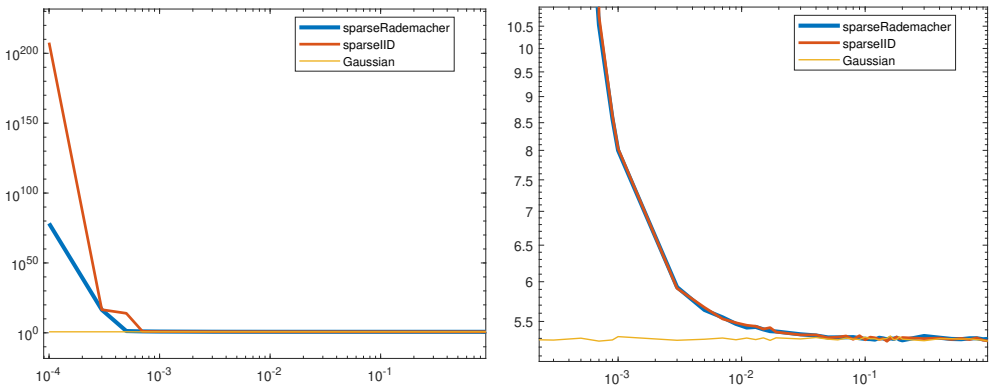

Fig. SM2: Condition number $\kappa(\mathbf{\Phi U})$ of $\mathbf{\Phi U}$ with respect to the sparsity level of SparseRademacher matrices. The right subfigure zooms out the left one in the interval $[10^{-3}, 1]$.

EXAMPLE SM3.1. *We set $n = 2000$, $d = 50$ and $s = 100$. The sparsity level ξ of sparse test matrices is set from $1e-4$ to 1 (including the 1% sparsity used in the main manuscript). All experiments are averaged by 1000 independent repeated experiments. We compare two types of sparse embeddings: (1) SparseRademacher matrix containing $nd \times \xi$ nonzero entries, the locations of nonzero entries are randomly selected, and the values of nonzero entries are independently set to $+1$ or -1 with probability 0.5. (2) SparseIID matrix [SM3], where each entry is independently set to $+1$ with probability $\xi/2$, -1 with probability $\xi/2$, and 0 with probability $1-\xi$. We also include the Gaussian embedding as a benchmark, where each entry is independently drawn from the standard normal distribution. We display the condition number $\kappa(\mathbf{\Phi U}) = \sigma_{\max}(\mathbf{\Phi U})/\sigma_{\min}(\mathbf{\Phi U})$ of $\mathbf{\Phi U}$ in Figure SM2.*

From Figure SM2, we can observe three phenomena:

(1) The subspace embedding property holds for sparseRademacher when the sparsity level is larger than $1e-3$. When the sparsity level decrease to less than $1e-3$, the condition numbers corresponding to the two test sparse embeddings increase significantly, especially for the sparseIID test matrix.

(2) When the sparsity level is larger than $1e-3$, the condition numbers corresponding to the SparseRademacher embedding are very close to that of the sparseIID embedding, while the latter is proved to satisfy the JL property [SM3].

(3) When the sparsity level is larger than $1e-2$, the condition numbers corresponding to the two sparse embeddings are relatively stable and close to that of the Gaussian embedding.

Based on the above observations, we find that when sparsity level is not too small (say, 10^{-3}), the SparseRademacher matrix may exhibit JL property in high-dimensional spaces in practice.

SM4. TYUC17 in Mixed-Precision and TYUC17-SPI Variant. At the end of Section 7 of the main manuscript, we have discussed that TYUC17 can also be implemented in mixed-precision, e.g., $\mathbf{W}$ in single-precision to allow a larger sketch while $\mathbf{Y}$ still in double-precision to avoid loss of orthogonality. A detailed algorithm is

Algorithm SM4.1 TYUC17-mixed-precision; see Section 7

Require: Single-precision sketch and test matrix $\mathbf{W} = \mathbf{\Psi A} \in \mathbb{R}^{d\times n}$, $\mathbf{\Psi} \in \mathbb{R}^{d\times m}$, double sketch $\mathbf{Y} = \mathbf{A\Omega} \in \mathbb{R}^{m\times s}$

Ensure: Rank-r approximation $\mathbf{U}\Sigma\mathbf{V}^T$ with orthonormal $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ and diagonal matrix $\Sigma \in \mathbb{R}^{r\times r}$

1: $[\mathbf{Q}, \sim] = \texttt{QR}(\mathbf{Q}, 0)$
2: $\mathbf{B} = (\mathbf{\Psi Q})\backslash\mathbf{W}$ ▷ Single-precision
3: $\mathbf{B} = \texttt{double}(\mathbf{B}) \in \mathbb{R}^{s\times n}$ ▷ Double-precision for avoiding loss of orthogonality in SVD. The space of $\mathbf{W}$ can be used for such a conversion if $d \geq 2s$; See Algorithm SM4.2.
4: $\mathbf{U}\Sigma\mathbf{V}^T = \texttt{SVD}(\mathbf{B}, r)$
5: $\mathbf{U} = \mathbf{QU}$
6: **return** $(\mathbf{U}, \Sigma, \mathbf{V})$

depicted in Algorithm SM4.1.

A TYUC17-SPI variant was introduced in Section 2.7. The variant generates two sketches only: $\mathbf{W} = \mathbf{\Psi A}$ and $\mathbf{Z} = \mathbf{A\Phi} \in \mathbb{R}^{m\times l}$. Then perform $\hat{\mathbf{Y}} = \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^q\tilde{\mathbf{\Omega}} \in \mathbb{R}^{m\times s}$ to obtain the rangefinder matrix. Here $\mathbf{\Psi} \in \mathbb{R}^{d\times m}, \mathbf{\Phi} \in \mathbb{R}^{m\times l}, \tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$ are independent random test matrices. $\mathbf{W}$ and $\mathbf{Z}$ are in single-precision. After the computation of $\hat{\mathbf{Y}}$, $\mathbf{Z}$ can be deleted, and its space can be used for converting $\hat{\mathbf{Y}}$ from single to double-precision. The remaining steps are the same as TYUC17-mixed-precision.

When $l \geq 2s$, the above conversion from single-precision to double-precision is possible. Without loss of generality assume that $\hat{\mathbf{Y}}$ is stored in the first s columns of the previous space of $\mathbf{Z}$ and is in the column-major format. The extra space for converting is also of size at least $m \cdot s$, just the remainig space of $\mathbf{Z}$. The procedure is depicted in Algorithm SM4.2. Note that this memory reuse is feasible in programming environments that allow for explicit memory management or fine-grained control over data allocation and deallocation.

Algorithm SM4.2 Convert single-precision matrix to double-precision

Require:
$\mathbf{Y}_{\text{single}}$: an $m \times s$ matrix in single precision (column-major)
T: extra space of size $m \cdot s + 2$

Ensure:
$\mathbf{Y}_{\text{double}}$ contains the double-precision copy of $\mathbf{Y}_{\text{single}}$

1: virtually view $\mathbf{Y}_{\text{single}}$ and T as a $(2ms + 2) \times 1$ vector $\mathbf{Z}$
2: **for** $k \leftarrow m \cdot s$ **downto** 1 **do**
3: $\quad\mathbf{Z}(2k + 1 : 2k + 2) \leftarrow \texttt{double}(Y_{\text{single}}(k))$
4: **end for**
5: **return Z**

Example SM4.1. *In this example, we compare TYUC17-mixed-precision (Algorithm SM4.1) and TYUC17-SPI variant in Section 2.7 (Algorithm SM4.3) on synthetic data as in Section 4.1.1 under the same storage budget. We present the oracle error only.*

Algorithm SM4.3 TYUC17-SPI variant; see Section 2.7

Require: Two single-precision sketches $\mathbf{Z} = \mathbf{A\Phi} \in \mathbb{R}^{m\times l}$, $\mathbf{W} = \mathbf{\Psi A} \in \mathbb{R}^{d\times n}$, single test matrix $\mathbf{\Psi} \in \mathbb{R}^{d\times m}$, double test matrix $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$

Ensure: Rank-r approximation $\mathbf{U\Sigma V}^T$ with orthonormal $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ and diagonal matrix $\Sigma \in \mathbb{R}^{r\times r}$

1: $\mathbf{M}_{\text{tmp}} = \texttt{double}(\mathbf{Z}^T\mathbf{Z}) \in \mathbb{R}^{l\times l}$ ▷ Only take l^2 extra storage for conversion if l is small.

2: $\hat{\mathbf{Y}} = \mathbf{ZM}_{\text{tmp}}^q\tilde{\mathbf{\Omega}} \in \mathbb{R}^{m\times s}$ ▷ Optional: re-orthonormalization on $\mathbf{M}_{\text{tmp}}\tilde{\mathbf{\Omega}}$

3: $[\mathbf{Q}, \sim] = \texttt{QR}(\texttt{double}(\tilde{\mathbf{Y}}), 0)$ ▷ Use the space of $\mathbf{Z}$ to double $\hat{\mathbf{Y}}$ to avoid loss of orthogonality in $\mathbf{Q}$. The remaining steps are the same as TYUC17-mixed-precision (Algorithm SM4.1)

4: $\mathbf{B} = (\mathbf{\Psi Q})\backslash\mathbf{W}$

5: $\mathbf{B} = \texttt{double}(\mathbf{B})$ ▷ The same as that in Algorithm SM4.1

6: $\mathbf{U\Sigma V}^T = \texttt{SVD}(\mathbf{B}, r)$

7: $\mathbf{U} = \mathbf{QU}$

8: **return** $(\mathbf{U}, \Sigma, \mathbf{V})$

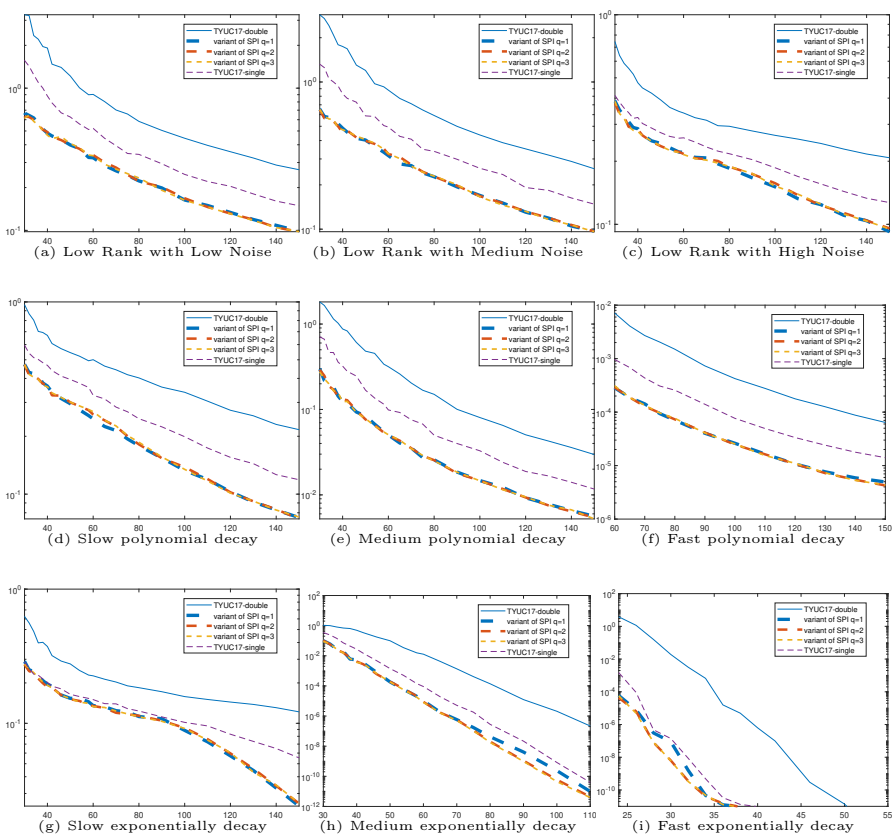

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Fig. SM3: Figures of Synectic data. X-axis means the T represented the storage budget. Y-axis means the relative Frobenius error S_F. All results are averaged by 20 independent repeated experiments.

Figure SM3 shows the performance of the prototype TYUC17 (denoted as TYUC17-double), TYUC17-mixed-precision (denoted as TYUC17-single), and TYUC17-SPI variant (denoted as SPI variant). Here TYUC17-double acts as a baseline. From the figure, we first observe that TYUC17-single outperforms TYUC17-double, in that TYUC17-single allows a larger $\mathbf{W}$, although it is in single-precision. Next, we can also see that TYUC17-SPI variant still outperforms TYUC17-single, thanks to the larger single-precision sketch $\mathbf{Z}$ to retain more range information of the data matrix. The improvement is more significant when the data matrix has a slow to moderate spectrum decay. The exception is that when the spectrum exhibits a fast exponential decay and a larger storage budget is available (in the bottom-right subfigure), both of TYUC17-single and TYUC17-SPI variant are outperformed by TYUC17-double. This is due to that the former two reach the precision limit of their mixed-precision strategy, where the round-off errors become dominant.

SM5. Probabilistic Spectral Norm Bound for TYUC17. Similar to Lemma 5.8 of the main manuscript, when $d \geq s+4$, we have that conditioned on $E = \{\mathbf{\Omega} | \, \|\mathbf{A} - P_{\mathbf{A\Omega}}\mathbf{A}\| < e_1, \|\mathbf{A} - P_{\mathbf{A\Omega}}\mathbf{A}\|_F < e_2\}$, with failure probability at most $e^{-u^2/2} + 2t^{-(d-s)}$, there holds

$$\|\mathbf{A} - \mathbf{QB}\| \leq \frac{e\sqrt{d}t}{d-s+1}e_2 + \left(1 + t\sqrt{\frac{3s}{d-s+1}} + u\frac{e\sqrt{d}t}{d-s+1}\right)e_1.$$

For any $\varrho \leq s-4$ and $t \geq 1, u > 0$, with failure probability at most $2t^{-(s-\varrho)} + e^{-u^2/2}$ there holds

$$\|(I - P_{\mathbf{A\Omega}})\mathbf{A}\|_F \leq \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}}\right)\tau_{\varrho+1}(\mathbf{A}) + ut \cdot \frac{e\sqrt{s}}{s-\varrho+1} \cdot \sigma_{\varrho+1}(\mathbf{A});$$

$$\|(I - P_{\mathbf{A\Omega}})\mathbf{A}\| \leq \left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}} + ut \cdot \frac{e\sqrt{s}}{s-\varrho+1}\right)\sigma_{\varrho+1}(\mathbf{A}) + t \cdot \frac{e\sqrt{s}}{s-\varrho+1}\tau_{\varrho+1}(\mathbf{A}).$$

Thus with failure probability at most $e^{-u^2/2} + 2t^{-(d-s)} + 4t^{-(s-\varrho)} + 2e^{-u^2/2}$, $\|\mathbf{A} - \mathbf{QB}\|$ with $\mathbf{Q}$ and $\mathbf{B}$ generated by TYUC17 is upper bounded by

$$\begin{aligned}
&\frac{e\sqrt{d}t}{d-s+1} \cdot \left[\left(1 + t \cdot \sqrt{\frac{3\varrho}{s-\varrho+1}}\right)\tau_{\varrho+1}(\mathbf{A}) + ut \cdot \frac{e\sqrt{s}}{s-\varrho+1} \cdot \sigma_{\varrho+1}(\mathbf{A})\right] \\
&\quad + \left(1 + t\sqrt{\frac{3s}{d-s+1}} + u\frac{e\sqrt{d}t}{d-s+1}\right)\left[\left(1 + t\sqrt{\frac{3\varrho}{s-\varrho+1}} + ut\frac{e\sqrt{s}}{s-\varrho+1}\right)\sigma_{\varrho+1}(\mathbf{A})\right. \\
&\qquad\qquad \left. + t \cdot \frac{e\sqrt{s}}{s-\varrho+1}\tau_{\varrho+1}(\mathbf{A})\right].
\end{aligned}$$

SM6. Proof of Oblique Projection Error Decomposition. We provide a proof for the oblique projection error decomposition stated in Sect. 4.3.1 of the main manuscript:

$$\left\|\mathbf{A} - \mathbf{U}\Sigma\mathbf{V}^T\right\|_p = \|(\mathbf{A} - \mathbf{UU}^T\mathbf{A}) + (\mathbf{U}\tilde{\mathbf{U}}^T(\mathbf{Q}^T\mathbf{A} - (\mathbf{\Psi Q})^\dagger\mathbf{\Psi A}))\|_p.$$

Proposition SM6.1. *For any column orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{m\times s}$ and matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, let $\tilde{\mathbf{U}}, \Sigma, \mathbf{V}$ be the rank-r truncated SVD of $\mathbf{B} = (\mathbf{\Psi Q})^\dagger\mathbf{\Psi A}$, $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$. Then the following holds:*

$$\left\|\mathbf{A} - \mathbf{U}\Sigma\mathbf{V}^T\right\| = \left\|(\mathbf{A} - \mathbf{UU}^T\mathbf{A}) + \mathbf{U}\tilde{\mathbf{U}}^T(\mathbf{Q}^T\mathbf{A} - (\mathbf{\Psi Q})^\dagger\mathbf{\Psi A})\right\|.$$

Proof. By the definitions of $\tilde{\mathbf{U}}, \Sigma, \mathbf{V}$, and $\mathbf{B}$,

$$\tilde{\mathbf{U}}^T(\mathbf{\Psi Q})^\dagger \mathbf{\Psi A} = \tilde{\mathbf{U}}^T \mathbf{B} = \Sigma \mathbf{V}^T.$$

and we obtain that

$$\begin{aligned}\left\|\mathbf{A} - \mathbf{U}\Sigma\mathbf{V}^T\right\| &= \left\|\mathbf{A} - \mathbf{U}\mathbf{U}^T\mathbf{A} + \mathbf{U}\mathbf{U}^T\mathbf{A} - \mathbf{U}\tilde{\mathbf{U}}^T(\mathbf{\Psi Q})^\dagger\mathbf{\Psi A}\right\| \\ &= \left\|(\mathbf{A} - \mathbf{U}\mathbf{U}^T\mathbf{A}) + \mathbf{U}\tilde{\mathbf{U}}^T(\mathbf{Q}^T\mathbf{A} - (\mathbf{\Psi Q})^\dagger\mathbf{\Psi A})\right\|. \qquad \square\end{aligned}$$

SM7. Application of SPI to the Streaming Algorithm TYUC19. In this section, we demonstrate that SPI can also be applied to other randomized low-rank approximation algorithms. Even without hardware specifications, we can still apply SPI to algorithms under memory-limited scenarios. For example, the streaming algorithm TYUC19 [SM4], proposed by Tropp et al. We first recall the algorithm in Section SM7.1, then incorporate SPI into TYUC19 in Section SM7.2, and perform numerical comparisons. In Section SM7.3, we present the proof idea for establishing the deviation bound for TYUC19-SPI.

SM7.1. TYUC19 algorithm. We briefly recall the TYUC19 algorithm here. It uses two sketches:

$$\mathbf{Y} = \mathbf{A\Omega} \in \mathbb{R}^{m\times s}, \quad \mathbf{X} = \mathbf{\Gamma A} \in \mathbb{R}^{s\times n}$$

to capture the projection space, and then applies two-sided sketching:

$$\mathbf{K} = \mathbf{\Phi A \Psi}^T \in \mathbb{R}^{d\times d},$$

to contains the fresh information, where $\mathbf{\Omega} \in \mathbb{R}^{n\times s}, \mathbf{\Gamma} \in \mathbb{R}^{s\times m}, \mathbf{\Phi} \in \mathbb{R}^{d\times m}$, and $\mathbf{\Psi} \in \mathbb{R}^{d\times n}$ are independent random test matrices.

The range and co-range of $\mathbf{A}$ are constructed by orthonormalizing the sketches $\mathbf{Y}$ and $\mathbf{X}$:

$$\mathbf{Q} = \texttt{QR}(\mathbf{Y}) \in \mathbb{R}^{m\times s}, \quad \mathbf{P} = \texttt{QR}(\mathbf{X}^T) \in \mathbb{R}^{n\times s}.$$

Then, an approximation of the "core matrix" is obtained by solving least squares problems:

$$\mathbf{C} = ((\mathbf{\Phi Q})\backslash\mathbf{K})/(\mathbf{\Psi P})^T \in \mathbb{R}^{s\times s},$$

which returns the approximate triple $(\mathbf{Q}, \mathbf{C}, \mathbf{P})$.

Further steps, such as truncated SVD, can be applied to $\mathbf{C}$ to obtain the final result, similar to TYUC17. The algorithm is shown in Algorithm SM7.1.

SM7.2. TYUC19-SPI. In the multi-level caches and distributed systems, SPI can be incorporated into TYUC19, as that for TYUC17 discussed in Section 2.5. The details is similar and we omit them. Here we mainly focus on the mixed-precision strategy, and we will integrate SPI into TYUC19 in a similar way as the TYUC17-SPI variant in Section 2.7 (Algorithm SM4.3). To be specific, we sketch two sketches

$$\mathbf{Z} = \mathbf{A\Omega} \in \mathbb{R}^{m\times l} \text{ and } \mathbf{W} = \mathbf{\Gamma A} \in \mathbb{R}^{l\times n}$$

Algorithm SM7.1 TYUC19 [SM4]

Require: Sketches $\mathbf{Y} \in \mathbb{R}^{m\times s}$, $\mathbf{X} \in \mathbb{R}^{s\times n}$, $\mathbf{K} \in \mathbb{R}^{d\times d}$, $\mathbf{\Phi} \in \mathbb{R}^{d\times m}$,$\mathbf{\Psi} \in \mathbb{R}^{n\times d}$.
Ensure: Rank-r approximation in the form $\mathbf{U}\Sigma\mathbf{V}^T$ with orthonormal $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ and diagonal $\Sigma \in \mathbb{R}^{r\times r}$

$[\mathbf{Q}, \sim] = \texttt{QR}(\mathbf{Y}, 0)$
$[\mathbf{P}, \sim] = \texttt{QR}(\mathbf{X}^T, 0)$
$\mathbf{C} = ((\mathbf{\Phi Q})^T \backslash \mathbf{K})/((\mathbf{\Psi P})^T)$
$[\mathbf{U}, \Sigma, \mathbf{V}] = \texttt{SVD}(\mathbf{C}, r)$
$\mathbf{U} = \mathbf{QU}$
$\mathbf{V} = \mathbf{PV}$
return $(\mathbf{U}, \Sigma, \mathbf{V})$

in single-precision, and sketch

$$\mathbf{K} = \mathbf{\Phi A \Psi}^T \in \mathbb{R}^{d\times d},$$

in double-precision as that in TYUC19. Then we generate two small random test matrices $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$ and $\tilde{\mathbf{\Gamma}} \in \mathbb{R}^{s\times l}$, and apply the SPI to get $\hat{\mathbf{Y}} \in \mathbb{R}^{m\times s}$ and $\hat{\mathbf{X}} \in \mathbb{R}^{s\times n}$ as follows:

$$\hat{\mathbf{Y}} = \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^q\tilde{\mathbf{\Omega}} \in \mathbb{R}^{m\times s}, \quad \hat{\mathbf{X}} = \tilde{\mathbf{\Gamma}}(\mathbf{W}\mathbf{W}^T)^q\mathbf{W} \in \mathbb{R}^{s\times n}.$$

As TYUC17-SPI variant, $\hat{\mathbf{Y}}$ can be stored in the first s columns of $\mathbf{Z}$, and $\hat{\mathbf{X}}$ can be stored in the first s rows of the previous space of $\mathbf{W}$. When $l \geq 2s$, we can use the space of $\mathbf{Z}$ and $\mathbf{W}$ to convert $\hat{\mathbf{Y}}$ and $\hat{\mathbf{X}}$ from single-precision to double-precision, using Algorithm SM4.2. We then orthogonal $\hat{\mathbf{Y}}$ and $\hat{\mathbf{X}}^T$ to obtain $\mathbf{Q}$ and $\mathbf{P}$. The remaining steps are the same as TYUC19. We depict the algorithm in Algorithm SM7.2.

Algorithm SM7.2 TYUC19-SPI

Require: Two single-precision sketches $\mathbf{Z} = \mathbf{A\Omega} \in \mathbb{R}^{m\times l}$, $\mathbf{W} = (\mathbf{\Gamma A})^T \in \mathbb{R}^{l\times n}$, double-precision sketch $\mathbf{K} = \mathbf{\Phi A \Psi} \in \mathbb{R}^{d\times d}$, test matrices $\mathbf{\Phi} \in \mathbb{R}^{d\times m}$,$\mathbf{\Psi} \in \mathbb{R}^{n\times d}$, $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{l\times s}$,$\tilde{\mathbf{\Gamma}} \in \mathbb{R}^{s\times l}$.
Ensure: Rank-r approximation in the form $\mathbf{U}\Sigma\mathbf{V}^T$ with orthonormal $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ and diagonal $\Sigma \in \mathbb{R}^{r\times r}$

$\mathbf{M}_{\text{tmp}} = \texttt{double}(\mathbf{Z}^T\mathbf{Z}) \in \mathbb{R}^{l\times l}$ ▷ Only take l^2 extra storage for conversion if l is small.
$\hat{\mathbf{Y}} = \mathbf{Z}\mathbf{M}_{\text{tmp}}^q\tilde{\mathbf{\Omega}} \in \mathbb{R}^{m\times s}$ ▷ Optional: re-orthonormalization on $\mathbf{M}_{\text{tmp}}\tilde{\mathbf{\Omega}}$
$[\mathbf{Q}, \sim] = \texttt{QR}(\texttt{double}(\hat{\mathbf{Y}}), 0)$ ▷ Double $\hat{\mathbf{Y}}$ to avoid loss of orthogonality in $\mathbf{Q}$
$\mathbf{N}_{\text{tmp}} = \texttt{double}(\mathbf{W}\mathbf{W}^T) \in \mathbb{R}^{l\times l}$
$\hat{\mathbf{X}} = \tilde{\mathbf{\Gamma}}\mathbf{N}_{\text{tmp}}^q\mathbf{W} \in \mathbb{R}^{s\times n}$ ▷ Optional: re-orthonormalization on $\mathbf{N}_{\text{tmp}}\tilde{\mathbf{\Gamma}}$
$[\mathbf{P}, \sim] = \texttt{QR}(\texttt{double}(\hat{\mathbf{X}}^T), 0)$ ▷ Double $\hat{\mathbf{X}}^T$ to avoid loss of orthogonality in $\mathbf{P}$
$\mathbf{C} = ((\mathbf{\Phi Q})^T \backslash \mathbf{K})/((\mathbf{\Psi P})^T)$ ▷ The remaining steps are the same as TYUC19 (Algorithm SM7.1)
$[\mathbf{U}, \Sigma, \mathbf{V}] = \texttt{SVD}(\mathbf{C}, r)$
$\mathbf{U} = \mathbf{QU}$
$\mathbf{V} = \mathbf{PV}$
return $(\mathbf{U}, \Sigma, \mathbf{V})$

EXAMPLE SM7.1. *In this example, we compare TYUC19 (Algorithm SM7.1) and TYUC19-SPI (Algorithm SM7.2) on synthetic data as in Section 4.1.1 under the same storage budget. We present the oracle error only.*

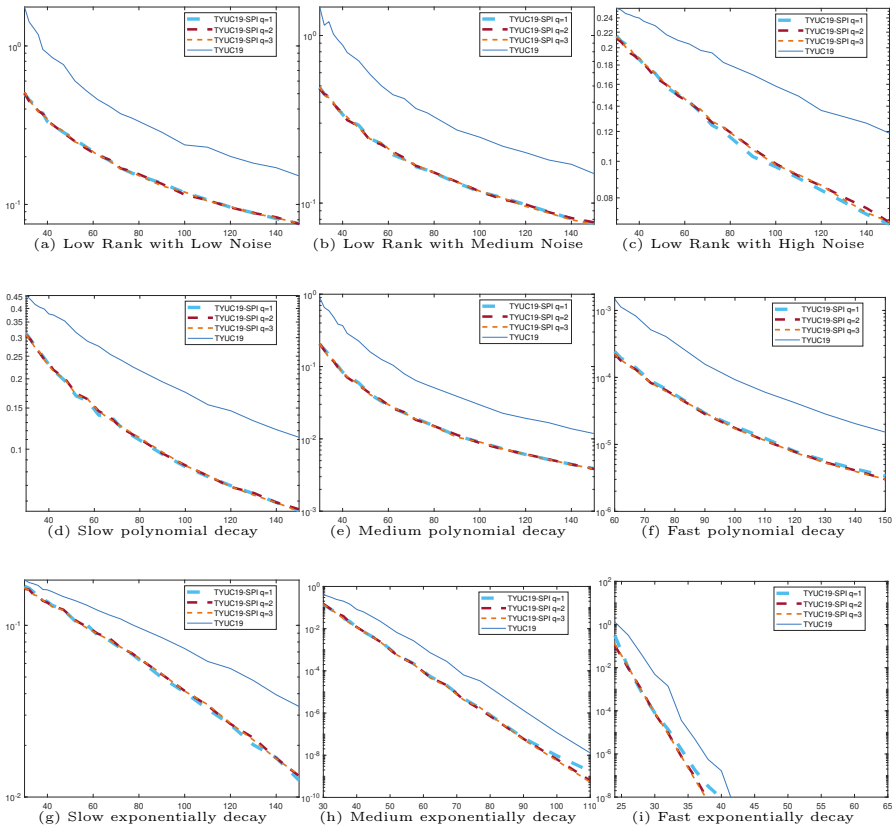

Fig. SM4: Figures of Synectic data. X-axis means the T represented the storage budget. Y-axis means the relative Frobenius error S_F. All results are averaged by 20 independent repeated experiments.

Figure SM4 shows the performance of TYUC19-SPI and TYUC19 under the same storage budget. We can observe that the trend is similar to that of TYUC17-SPI. In summary, SPI can be not only integrated into TYUC17, but also into TYUC19, and possibly other one-pass algorithms, to improve their accuracy, especially when the decay of the data matrix is flat or moderate.

SM7.3. Analysis of deviation bound for TYUC19-SPI. We can also obtain a deviation bound for the TYUC19-SPI. Here for notational simplicity, we do not present the final bound; instead we provide the proof idea on how to obtain the final bound.

Let $\hat{\mathbf{A}} = \mathbf{QCP}^T$, where $\mathbf{Q}, \mathbf{C}, \mathbf{P}$ are generated in Steps 3, 6, and 7 of Algorithm SM7.2. We first follow [SM4, Section SM1.6; Lemma SM1.5] to make a decompose:

$$\begin{aligned}
\left\|\mathbf{A} - \hat{\mathbf{A}}\right\|_F^2 &= \left\|\mathbf{A} - \mathbf{QCP}^T\right\|_F^2 \\
&= \left\|\mathbf{A} - \mathbf{QQ}^T\mathbf{APP}^T + \mathbf{Q}\left(\mathbf{Q}^T\mathbf{AP} - \mathbf{C}\right)\mathbf{P}^T\right\|_F^2 \\
&= \left\|\mathbf{A} - \mathbf{QQ^TAPP^T}\right\|_F^2 + \left\|\mathbf{Q}^T\mathbf{AP} - \mathbf{C}\right\|_F^2,
\end{aligned}$$

where the equality only uses the orthonormality of $\mathbf{Q}$. Further denote

$$\mathbf{P}_\perp \mathbf{P}_\perp^T = \mathbf{I} - \mathbf{P}\mathbf{P}^T, \quad \mathbf{Q}_\perp \mathbf{Q}_\perp^T = \mathbf{I} - \mathbf{Q}\mathbf{Q}^T,$$

where $\mathbf{P}_\perp \in \mathbb{R}^{n\times(n-s)}$, $\mathbf{Q}_\perp \in \mathbb{R}^{m\times(m-s)}$. Note that

$$\begin{aligned}\left\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^{\mathbf{T}}\mathbf{A}\mathbf{P}\mathbf{P}^{\mathbf{T}}\right\|_F^2 &= \left\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^{\mathbf{T}}\mathbf{A}\mathbf{P}\mathbf{P}^{\mathbf{T}} + \mathbf{Q}\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp\mathbf{P}_\perp^T - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp\mathbf{P}_\perp^T\right\|_F^2 \\ &= \left\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^{\mathbf{T}}\mathbf{A}^{\mathbf{T}} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp\mathbf{P}_\perp^T\right\|_F^2 \\ &= \left\|\left(\mathbf{I} - \mathbf{Q}\mathbf{Q}^T\right)\mathbf{A}\right\|_F^2 + \left\|\mathbf{A}\left(\mathbf{I} - \mathbf{P}\mathbf{P}^T\right)\right\|_F^2.\end{aligned}$$

Thus

$$\left\|\mathbf{A} - \hat{\mathbf{A}}\right\|_F^2 = \left\|\left(\mathbf{I} - \mathbf{Q}\mathbf{Q}^T\right)\mathbf{A}\right\|_F^2 + \left\|\mathbf{A}\left(\mathbf{I} - \mathbf{P}\mathbf{P}^T\right)\right\|_F^2 + \left\|\mathbf{Q}^T\mathbf{A}\mathbf{P} - \mathbf{C}\right\|_F^2. \tag{SM7.1}$$

The deviation bound for the first two terms have been obtained in Theorem 6.7 of the main manuscript. The expectation bound of $\left\|\mathbf{Q}^T\mathbf{A}\mathbf{P} - \mathbf{C}\right\|_F^2$ has been established in [SM4, Lemma SM1.4]. Here we need a deviation bound. [SM4, Lemma SM1.3] shows the decomposition:

$$\mathbf{C} - \mathbf{Q}^T\mathbf{A}\mathbf{P} = \mathbf{\Phi}_1^\dagger\mathbf{\Phi}_2(\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}) + (\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp)\mathbf{\Psi}_2^T(\mathbf{\Psi}_1^\dagger)^T + \mathbf{\Phi}_1^\dagger\mathbf{\Phi}_2(\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}_\perp)\mathbf{\Psi}_2^T(\mathbf{\Psi}_1^\dagger)^T.$$

Here

$$\begin{aligned}\mathbf{\Phi}_1 &= \mathbf{\Phi}\mathbf{Q} \in \mathbb{R}^{d\times s}, \quad \mathbf{\Phi}_2 = \mathbf{\Phi}\mathbf{Q}_\perp \in \mathbb{R}^{d\times(m-s)}, \\ \mathbf{\Psi}_1 &= \mathbf{\Psi}\mathbf{P} \in \mathbb{R}^{d\times s}, \quad \mathbf{\Psi}_2 = \mathbf{\Psi}\mathbf{P}_\perp \in \mathbb{R}^{d\times(n-s)}.\end{aligned}$$

Then we simply decompose $\left\|\mathbf{C} - \mathbf{Q}^T\mathbf{A}\mathbf{P}\right\|_F^2$ as

$$\begin{aligned}\left\|\mathbf{C} - \mathbf{Q}^T\mathbf{A}\mathbf{P}\right\|_F^2 \le{}& 3\left\|\mathbf{\Phi}_1^\dagger\mathbf{\Phi}_2(\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P})\right\|_F^2 + 3\left\|(\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp)\mathbf{\Psi}_2^T(\mathbf{\Psi}_1^\dagger)^T\right\|_F^2 \\ &+ 3\left\|\mathbf{\Phi}_1^\dagger\mathbf{\Phi}_2(\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}_\perp)\mathbf{\Psi}_2^T(\mathbf{\Psi}_1^\dagger)^T\right\|_F^2.\end{aligned}$$

Similar to the arguments in Lemma 6.8 of the main manuscript we can obtain that

$$\begin{aligned}&\mathbb{P}\left(\left\|\mathbf{\Phi}_1^\dagger\mathbf{\Phi}_2(\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P})\right\|_F^2 > \alpha\left\|\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}\right\|_F^2 t^2(1+\mu)\right) \le e^{-2\mu^2/2} + t^{-(d-s)}; \\ &\mathbb{P}\left(\left\|(\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp)\mathbf{\Psi}_2^T(\mathbf{\Psi}_1^\dagger)^T\right\|_F^2 > \alpha\left\|\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp\right\|_F^2 t^2(1+\mu)\right) \le e^{-2\mu^2/2} + t^{-(d-s)}; \\ &\mathbb{P}\left(\left\|\mathbf{\Phi}_1^\dagger\mathbf{\Phi}_2(\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}_\perp)\mathbf{\Psi}_2^T(\mathbf{\Psi}_1^\dagger)^T\right\|_F^2 > \alpha^2\left\|\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}_\perp\right\|_F^2 t^2(1+\mu)\right) \le 2e^{-2\mu^2/2} + 2t^{-(d-s)},\end{aligned}$$

where $\alpha = \frac{3s}{d-s+1}$. Combining the above yields

$$\begin{aligned}\left\|\mathbf{C} - \mathbf{Q}^T\mathbf{A}\mathbf{P}\right\|_F^2 \le{}& 3\alpha t^2(1+\mu)\left(\left\|\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}\right\|_F^2 + \left\|\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp\right\|_F^2\right) \\ &+ 3\alpha^2 t^2(1+\mu)\left\|\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}_\perp\right\|_F^2\end{aligned}$$

with failure probability $4e^{-2\mu^2/2} + 2t^{-(d-s)}$.

Note that $\left\|\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}\right\|_F^2 + \left\|\mathbf{Q}^T\mathbf{A}\mathbf{P}_\perp\right\|_F^2 + \left\|\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}_\perp\right\|_F^2 = \left\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\mathbf{P}\mathbf{P}^T\right\|_F^2$. Thus the above becomes

$$\begin{aligned}\left\|\mathbf{C} - \mathbf{Q}^T\mathbf{A}\mathbf{P}\right\|_F^2 \leq 3\alpha t^2(1+\mu)\left\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\mathbf{P}\mathbf{P}^T\right\|_F^2 \\ + 3(\alpha^2-\alpha)t^2(1+\mu)\left\|\mathbf{Q}_\perp^T\mathbf{A}\mathbf{P}_\perp\right\|_F^2\end{aligned}$$

with failure probability $4e^{-2\mu^2/2} + 2t^{-(d-s)}$. When $d > 4s$ the last term becomes non-positive. As a result, when $d > 4s$ and conditioned on $\mathbf{P}, \mathbf{Q}$, the above relation together with (SM7.1) yields

$$\left\|\mathbf{A} - \hat{\mathbf{A}}\right\|_F^2 \leq \left(1 + 3\alpha t^2(1+\mu)\right)\left(\left\|\left(\mathbf{I} - \mathbf{Q}\mathbf{Q}^T\right)\mathbf{A}\right\|_F^2 + \left\|\mathbf{A}\left(\mathbf{I} - \mathbf{P}\mathbf{P}^T\right)\right\|_F^2\right)$$

with failure probability $4e^{-2\mu^2/2} + 2t^{-(d-s)}$. This in connection with our previous estimations of $\left\|\mathbf{A} - \mathbf{Q}\mathbf{Q}^T\mathbf{A}\right\|_F$ in Theorem 6.7 of the main manuscript yields the final deviation bound for the TYUC19-SPI. We omit the detailed results.

REFERENCES

[SM1] A. Kireeva and J. A. Tropp, *Randomized matrix computations: Themes and variations*, arXiv:2402.17873, (2024).

[SM2] P.-G. Martinsson and J. A. Tropp, *Randomized numerical linear algebra: foundations and algorithms*, Acta Numer., 29 (2020), pp. 403–572.

[SM3] J. A. Tropp, *Comparison theorems for the minimum eigenvalue of a random positive-semidefinite matrix*, 2025, https://arxiv.org/abs/2501.16578.

[SM4] J. A. Tropp, A. Yurtsever, M. Udell, and V. Cevher, *Streaming low-rank matrix approximation with an application to scientific simulation*, SIAM J. Sci. Comput., 41 (2019), pp. A2430–A2463.